

Homework 1

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1.

a.

$$\frac{\frac{\sigma' = \sigma[x \mapsto -1]}{\langle \sigma, x := -1; x \rangle \longrightarrow \langle \sigma', x \rangle} \text{ ASSGN}}{\frac{\langle \sigma, \text{clamp}((x := -1; x), (x := 2; x)) \rangle \longrightarrow \langle \sigma', \text{clamp}(x, (x := 2; x)) \rangle}{\langle \sigma, \text{clamp}((x := -1; x), (x := 2; x)) + x \rangle \longrightarrow \langle \sigma', \text{clamp}(x, (x := 2; x)) + x \rangle} \text{ CLAMP1}} \text{ LADD}$$

b.

$$\begin{aligned} & \langle \sigma, \text{clamp}((x := -1; x), (x := 2; x)) + x \rangle \\ & \longrightarrow \langle \sigma', \text{clamp}(x, (x := 2; x)) + x \rangle \\ & \longrightarrow \langle \sigma', \text{clamp}(-1, (x := 2; x)) + x \rangle \\ & \longrightarrow \langle \sigma'', \text{clamp}(-1, x) + x \rangle \\ & \longrightarrow \langle \sigma'', \text{clamp}(-1, 2) + x \rangle \\ & \longrightarrow \langle \sigma'', 0 + x \rangle \\ & \longrightarrow \langle \sigma'', 0 + 2 \rangle \\ & \longrightarrow \langle \sigma'', 2 \rangle \end{aligned}$$

c.

Update the following rules

$$\frac{n_1 < 0}{\langle \sigma, \text{clamp}(n_1, e_2) \rangle \longrightarrow \langle \sigma, 0 \rangle} \text{ CLAMPLNEG}$$

$$\frac{n_1 \geq 0 \quad \langle \sigma, e_2 \rangle \longrightarrow \langle \sigma', e'_2 \rangle}{\langle \sigma, \text{clamp}(n_1, e_2) \rangle \longrightarrow \langle \sigma', \text{clamp}(n_1, e'_2) \rangle} \text{ CLAMP2}$$

Also, $n_1 \geq 0$ can be removed as a condition for CLAMP2NEG.

2.

a.

No, for example $\langle \sigma, (1+2)/(3+4) \rangle$ can step to $\langle \sigma, 3/(3+4) \rangle$ using LDIV, or to $\langle \sigma, (1+2)/7 \rangle$ using RDIV.

b.

Yes, because the initial language terminates (either returns an *Int* or is blocked), so the evaluation of e_1 and e_2 in the division either gets blocked (terminates) or returns two *Ints* n_1 and n_2 . At this point, if $n_2 = 0$ the execution terminates, and if $n_2 \neq 0$ an *Int* is returned and the execution terminates.

c.

No, for example $\langle \emptyset, 1/0 \rangle \not\rightarrow$.

3.

a.

$$\frac{}{\varepsilon \in Str} \quad \frac{}{c \in Str} \quad \frac{s_1 \in Str \quad s_2 \in Str}{s_1 \cdot s_2 \in Str}$$

b.

We want to prove $\forall s \in Str. P(s)$. By structural induction on s :

- **Case** ε : Prove $P(\varepsilon)$.
- **Case** c : Prove $P(c)$.
- **Case** $s_1 \cdot s_2$: Prove $P(s_1) \wedge P(s_2) \implies P(s_1 \cdot s_2)$.

c.

$$\frac{}{((c_1, c), c_1 \cdot c) \in append} \quad \frac{}{((s_1 \cdot s_2, c), (s_1 \cdot s_2) \cdot c) \in append}$$

d.

$$\frac{}{(c, c) \in last} \quad \frac{(s_2, c) \in last}{(s_1 \cdot s_2, c) \in last}$$

Generative AI disclosure

Claude Sonnet was used to translate handwriting to LaTeX. AI was not used in any other way for this homework.

Debriefing

1. 1h
2. Easy
3. Yes
4. 90%